

ADP AI - Prediction Report

Generated by ADP AI Random Forest Prediction Pipeline

Report ID: fec70bfb-0f33-450b-86f8-dc93d75c4c55

Algorithm: Random Forest Regressor

Model Type: regression

Accuracy Score: 99.7%

Dataset Details

Target Column: Data_value

Total Rows: 9555

Total Columns: 14

Model Selection

AI Detected Intent: auto

Final Task Type: regression

Decision Reason: The target column is numeric and continuous, so ADP AI selected Random Forest Regressor.

Metrics

R2 Score: 0.997

Mean Absolute Error: 179.6446

Root Mean Squared Error: 446.2558

Training Rows: 7608

Testing Rows: 1902

Top Feature Importance

Series_reference: 0.580538

Series_title_2: 0.376894

Period: 0.03721

Series_title_1: 0.001848

Group: 0.001369

Series_title_3: 0.001327

STATUS: 0.000639

Series_title_4: 0.000175

Suppressed: 0.0

UNITS: 0.0

Magnitude: 0.0

Subject: 0.0

Series_title_5: 0.0

Prediction Samples

Actual: 23883.359 | Predicted: 23883.7738

Actual: 666.243 | Predicted: 672.3676

Actual: 378.916 | Predicted: 378.8055

Actual: 292.64 | Predicted: 408.5709

Actual: 11396.519 | Predicted: 11536.1925

Actual: 1548.216 | Predicted: 1638.1659

Actual: 917.918 | Predicted: 835.2046

Actual: 1225.711 | Predicted: 1192.6614

Actual: 4088.925 | Predicted: 4024.1008

Actual: 3684.742 | Predicted: 3684.703

Smart Recommendations

1. Data quality: 14.31% values are missing. Fill important missing values before final prediction.
2. No duplicate rows found, so the dataset is suitable for prediction analysis.
3. Target analysis: Average Data_value is 5515.75, median is 2374.1, minimum is -398.19, and maximum is 43002.43.
4. Best performing group in 'Series_reference' is 'BDCQ.SF1FF1CT' with average Data_value of 32234.01. Follow patterns from this group.
5. Best performing group in 'Series_title_2' is 'Wholesale Trade' with average Data_value of 21468.24. Follow patterns from this group.
6. High priority: Improve 'Period' because it has a positive relationship with Data_value. Random Forest importance is 3.72%.
7. Best performing group in 'Series_title_1' is 'Sales (operating income)' with average Data_value of 8275.59. Follow patterns from this group.
8. Best performing group in 'Group' is 'Industry by financial variable (NZSIOC Level 1)' with average Data_value of 7876.31. Follow patterns from this group.
9. Improvement plan: Focus on the top important features that influence 'Data_value' because they have the strongest impact on prediction.

Prediction Flow

1. User uploaded dataset
2. ADP AI read rows and columns
3. ADP AI checked target column
4. ADP AI cleaned missing values
5. ADP AI prepared numeric and categorical features
6. ADP AI selected suitable ML model
7. Random Forest model trained using train-test split
8. Model predictions generated
9. Evaluation metrics calculated
10. Feature importance generated
11. Recommendation intelligence generated
12. Prediction report prepared